

Resolving Availability and Run-time Integrity Conflicts in Real-Time Embedded Systems

Adam Ilyas Caulfield*, Muhammad Wasif Kamran*, N. Asokan*,⁺

^{*}University of Waterloo, ⁺KTH Royal Institute of Technology

acaulfield@uwaterloo.ca, mwkamran@uwaterloo.ca, asokan@acm.org

Abstract

Run-time integrity enforcement in real-time systems presents a fundamental conflict with availability. Existing approaches in real-time systems primarily focus on minimizing the run-time overhead of monitoring. After a violation is detected, prior works face a trade-off: (1) prioritize availability and allow a compromised system to continue to ensure applications meet their deadlines, or (2) prioritize security by generating a fault to abort all execution.

In this work, we propose PAIR, an approach that offers a middle ground between the stark extremes of this trade-off. PAIR monitors real-time tasks for run-time integrity violations and maintains an *Availability Region (AR)* of all tasks that are *safe to continue*. When a task causes a violation, PAIR triggers a non-maskable interrupt to kill the task and continue executing a non-violating task within AR. Thus, PAIR ensures only violating tasks are prevented from execution, while granting availability to remaining tasks. With its hardware approach, PAIR does not cause any run-time overhead to the executing tasks. PAIR’s design is formally verified, integrates with real-time operating systems (RTOSs), and is affordable to low-end microcontroller units (MCUs) by incurring +2.3% overhead in memory and hardware usage.

1 Introduction

Embedded systems have become ubiquitous, often serving important roles in critical infrastructure responsible for real-time sensing or actuation, where deterministic and timely responses are crucial for functionality and safety. They are implemented with microcontroller units (MCUs), low-powered resource-constrained processing units optimized for energy efficiency. MCUs typically lack advanced features, such as memory management units (MMUs) or the ability to run rich operating systems (OS) that facilitate inter-process isolation. To make up for this, real-time operating systems (RTOSs) can be incorporated as an additional software layer to provide basic task management (e.g., scheduling, inter-task communication,

timing services) with low overhead. Still, limited architectural protections make MCUs vulnerable [43].

One prevalent threat includes run-time attacks, in which an adversary ($\mathcal{A}_{dv}$) could inject code into memory, then exploit a vulnerability (e.g., a buffer overflow [18]) to redirect execution to the injected code. Although memory-protection techniques can mitigate code-injection-based control flow attacks, they do not protect against code-reuse attacks like return- or jump-oriented programming (ROP/JOP) [9, 53], that chain together carefully selected instruction gadgets from unmodified code. To account for such attacks, run-time integrity mechanisms are required.

A prominent run-time integrity mechanism is Control Flow Integrity (CFI) [10], which employs reference monitors to detect when control flow instructions (e.g., calls, returns, jumps) deviate from an expected control flow path. Recent CFI methods for embedded and real-time systems [42] enable cost-effective enforcement of valid control flows. Several designs [16, 33, 56, 57, 62, 63] focus on performing CFI checks in a way that minimizes run-time overhead on a program (or *task*) while monitoring integrity. However, most works overlook the handling of tasks *after* a CFI violation has been raised. On the other hand, availability-focused security techniques for MCUs [1–3, 12, 14, 21] do not account for run-time violations [2, 3, 21] or delay intervention until task completion [1] or verification [12, 14, 38]. Given this trade-off, it is unclear from current works how to achieve a middle ground between post-violation integrity and availability.

To bridge this gap, we introduce PAIR: *Preserving Availability and Integrity at Run-time*. PAIR demonstrates that both availability and run-time integrity can be prioritized simultaneously. PAIR hardware components monitor run-time integrity of independent tasks within a given program. Then, PAIR dynamically adds and removes them from an Availability Region (AR) based on their integrity status. When any run-time integrity violation is detected, PAIR aborts execution of the violating task, removes it from AR, and continues executing another task in AR. PAIR does not allow a task to return to AR until a trusted software update occurs. To summarize, we

claim the following contributions:

- PAIR, an architecture operating alongside an RTOS that maintains availability and integrity of benign tasks after integrity violations due to other tasks (Section 3.2);
- formally defined properties of PAIR security, its modules sub-properties, and demonstration via model-checking that security goals are upheld (Sections 4 and 5);
- design and implementation of PAIR hardware submodules as formally-verified finite state machines (FSMs) that implement the outlined sub-properties (Section 6);
- a FPGA prototype of PAIR, evaluated with a real-world RTOS and example benchmark applications, demonstrating applicability to low-end MCUs due to minimal memory and hardware overheads (Section 7);
- a response-time analysis of PAIR’s task revocation to understand its real-time implications by evaluating the conditions under which PAIR’s behavior maintains performance of non-violating tasks (Section 8);
- a case study of a simulated real-world real-time system executing atop PAIR, showing that PAIR effectively revokes access to violating tasks while maintaining core functionality of remaining tasks (Section 9).

2 Background

2.1 Control Flow Integrity (CFI)

CFI is a countermeasure for control flow hijacking attacks, which exploit memory vulnerabilities to launch unintended run-time behavior by overwriting *control data* (e.g., return addresses, indirect jump targets, function pointers). For example, an exploited buffer overflow can be used by $\mathcal{Adv}$ to perform ROP [53], in which $\mathcal{Adv}$ overwrites function return addresses to stitch together program gadgets to execute arbitrary (potentially malicious) behavior without corrupting instructions themselves and only corrupting return addresses. This can similarly be completed by corrupting *forward-edge* control data, such as function pointers or indirect jump targets, to stitch together gadgets in the form of JOP [9].

CFI defends against such attacks by restricting the targets of control flow instructions (e.g., returns, calls, jumps) that use the control data that are exploited during control flow hijacking attacks, like ROP/JOP. These instructions are restricted such that the target control data determining the destination of the control flow transfers is limited to a set of predetermined valid destinations. Installing CFI into a system requires two components: (1) valid destination sets for each control flow instruction (at either coarse- or fine-granularity), and (2) a reference monitor to detect violations and respond accordingly [5].

Many techniques employ their reference monitor through software instrumentation [4, 24, 28, 32, 33, 35, 36, 40, 49, 62, 66], enabling deployment of CFI without hardware changes. In these works, software instrumentation either installs an in-line

reference monitor which compares the control flow target to the valid set of destinations, or redirects to a trusted software module that implements the reference monitor. Despite being hardware-agnostic, many approaches based on software instrumentation assume the presence of an MMU and a rich OS to ensure the instrumentation is not tampered with at run-time.

As these system mechanisms typically are not present on embedded systems, CFI mechanisms typically employ some type of hardware support, either in the form of custom hardware extensions [20, 64] or by leveraging architectural features [4, 49, 57, 63, 66]. For example, features such as Branch Target Identification (BTI) and Pointer Authentication (PA) in ARM [8] have led to subsequent proposals for fully fledged CFI schemes [7, 39]. ARM has made BTI and PA available among their class of embedded devices, along with hardware tracing extensions that have also been leveraged to deploy CFI in embedded systems [57, 63].

Across particular approaches, one state-of-the-art approach for CFI is through implementing a shadow stack [11], a protected stack which is updated by the CFI reference monitor on each `call` instruction with the expected return address. Upon return, the CFI mechanism will compare the value that was popped from the normal stack to the expected value according to the shadow stack, and raise a violation when they do not match. This enables a fine-grained backward-edge defense, since each return address has exactly one target. Methods based on shadow stacks are effective across the device spectrum, and have been implemented in embedded systems [26, 62, 66].

2.2 Real-Time Operations in MCUs

For real-time operations, MCU software can use an RTOS for critical services (e.g., task scheduling, inter-task communication, timing adherence). After an application assigns tasks priority and schedules them through an RTOS, the RTOS manages their execution. To do so, the RTOS contains functions to control execution, such as killing a task or yielding to another task. Tasks run until they voluntarily yield or until a time period has elapsed, at which point the RTOS performs a context switch and resumes the next highest-priority task. Many works have proposed security mechanisms that account for real-time operations in embedded systems, including mechanisms to provide authenticated [13, 44, 45] and audited [12, 14, 38] proofs of valid execution, availability mechanisms for trusted software [2, 3, 21], and local run-time integrity in the form of CFI [42].

CFI for real-time embedded systems is either fully software-based [1, 56, 62] or uses hardware extensions with minimal software support [33, 57, 63]. Regardless of the approach, all aim to minimize disruptions to the executing task during the CFI monitoring. Upon the violation, they abort execution to prioritize run-time integrity, compromising the availability of a system in an unknown state. However, in the case of real-

time systems, aborting execution system-wide can be costly. An alternative approach addresses this trade-off by serving as a *passive protection system* by logging the violation for later handling [1]. However, this is equally unacceptable in this scenario, as it allows the hijacked task to continue with no guarantees regarding safety.

2.3 Linear Temporal Logic

We use Linear Temporal Logic (LTL) [60] to specify a formal model of PAIR hardware and the properties that it should satisfy. As shown in prior works [3, 13, 47, 48, 59], LTL specifications can be used to describe finite state machines implemented within hardware and to reason about their properties.

In particular, we define security properties and hardware submodules using LTL statements containing the following propositional connectives:

- conjunctions (e.g., logical AND) between two statements ($\wedge$);
- disjunctions (e.g., logical OR) between two statements ($\vee$);
- and implications between statements ($\rightarrow$).

Additionally, we use the following temporal connectives between system states:

- $X\phi$: denoting a *next* state condition (e.g., condition ϕ is true in the next state);
- $G\phi$: denoting a *global* state condition (e.g., ϕ globally holds for all future states).

In Section 7, we use model checking (NuSMV [17]) to determine whether the system adheres to the specified properties.

2.4 Scope

We consider embedded systems operating real-time or sensing-based tasks atop low-end, single-core MCUs, with limited addressable memory (e.g., 8-256KB), split into program memory (PMEM) and data memory (DMEM). They are typically 8- or 16-bit CPUs running at 1-16MHz (e.g., TI MSP430). Such devices usually run software at *bare metal*, meaning they lack MMUs, and thus perform memory accesses and execute instructions from physical addresses.

We consider the lowest end of MCUs, on which hardware features for access control [29, 41] are either limited or non-existent. For the sake of this work, we assume the latter. However, we do assume the MCU has an integrity monitor (e.g., CFI support) as this is widely studied and already deployed into real-world MCUs [8]. We consider implications of more advanced platforms in Section 11.

In terms of software, we assume modularized set of *tasks* execute on the MCU. Each task has differences in timing requirements and priority. To manage these task characteristics, we assume an RTOS is also in use for scheduling, timing services, and synchronization between the tasks.

3 Problem Statement

3.1 Motivating Example

Consider a simple MCU that is used to implement a syringe pump controller that allows physicians to monitor and deliver medication to patients remotely over the network [31, 54]. In this scenario, the device may be installed with multiple software submodules serving different roles, operating as separate tasks atop an RTOS for prioritization, scheduling, and timely execution. For example, it may include the following:

- a WiFi module for network communication,
- LCD display and buzzer control for user interface and feedback,
- sensor monitoring of patient vitals (e.g., temperature and heart-rate monitoring), and
- motor control for accurate and timely dosage injection.

Suppose the network communication task contains a buffer overflow [18] on the stack, enabling a control flow hijack attack (e.g., ROP [53]). Given a CFI integrity monitor has been installed in the device, attempts to exploit this vulnerability will be detected. As discussed in Section 2, current approaches would not offer a suitable balance: they would either suspending all tasks for the duration of the reset and recovery window, including motor control and sensor monitoring, or allow the compromised network communication module to continue without safety guarantees.

Scenarios like this demand a mechanism that can simultaneously prevent violating tasks while ensuring non-violating tasks can continue their time-sensitive execution. Critically, hardware-based mechanisms are required, as other software (e.g., tasks or the RTOS itself) may attempt to reschedule the violating task before it is properly remediated. We revisit this example as a case study in Section 9.

3.2 PAIR at a High Level

Unlike a typical system in which a run-time integrity monitor (IM) would directly invoke a system reset signal, PAIR's goal is to ensure non-violating tasks can continue safely after a run-time integrity violation has been detected. Therefore, upon receiving a redirected violation signal, PAIR can perform operations to ensure benign tasks can securely continue. PAIR goals can be reduced to upholding the following two security properties at run-time:

[P1] Post-Violation Availability: when a task has a run-time integrity violation, PAIR always (1) removes the task from AR and (2) triggers the yielding to a non-violating task;

[P2] Post-Violation Integrity: tasks outside AR can never re-execute until a software update has occurred.

A high-level overview of an PAIR-enabled system is depicted in Figure 1. It assumes a system that operates real-time tasks (denoted T_i) with an RTOS, while PAIR introduces two trusted software components. Initially, the MCU runs with no

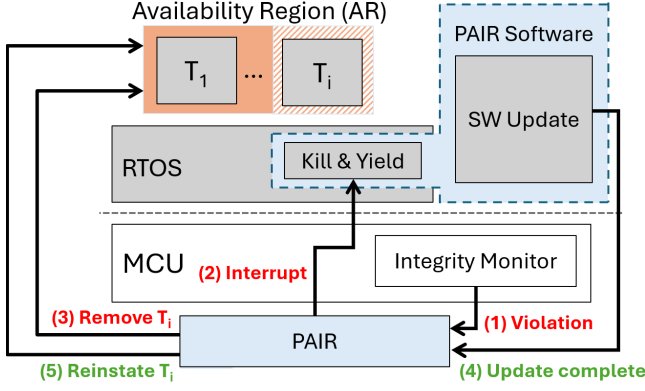


Figure 1: PAIR Overview: Tasks and an RTOS execute in software, with all tasks initially in the AR. After detecting an integrity violation (1), PAIR triggers the trampoline into RTOS to *kill-and-yield* (2) and removes the violating task from AR (3). Only after a trusted software update has completed (4) will PAIR reinstate the violating task into AR (5).

violations, and PAIR has assigned all tasks to AR. PAIR monitors MCU signals and memory to determine which real-time task is executing at any given moment. PAIR interfaces with an IM to detect the violation when it occurs (1). Upon this detection, PAIR will generate a non-maskable interrupt (2) to invoke the *kill-and-yield* trampoline, while also removing the violating task from AR. These steps are repeated for any other tasks that generate a violation. The bounds of AR are not expanded until a trusted software update has completed (4), and at this point PAIR will reinstate all violating tasks back into AR.

4 Adversary & System Models

4.1 Adversary Model

We consider $\mathcal{A}_{dv}$ that exploits software vulnerabilities to manipulate any data or software (tasks or the RTOS), unless it is explicitly protected by hardware. $\mathcal{A}_{dv}$ aims to cause run-time integrity violations (e.g., control flow hijack and ROP/JOP attacks [9, 53]) to invoke malicious actions or avoid critical actions. $\mathcal{A}_{dv}$ aims to continue its malicious actions even after detection by re-entering revoked tasks. Physical attacks to circumvent hardware protections are out of scope for this work, as they require orthogonal physical access control measures [50].

4.2 System Model

Figure 2 shows the assumed initial hardware configuration and memory layout. PAIR interfaces with the MCU to read the following signals pertaining to its run-time behavior:

- PC : the program counter register storing the memory address of the instruction that is executing;

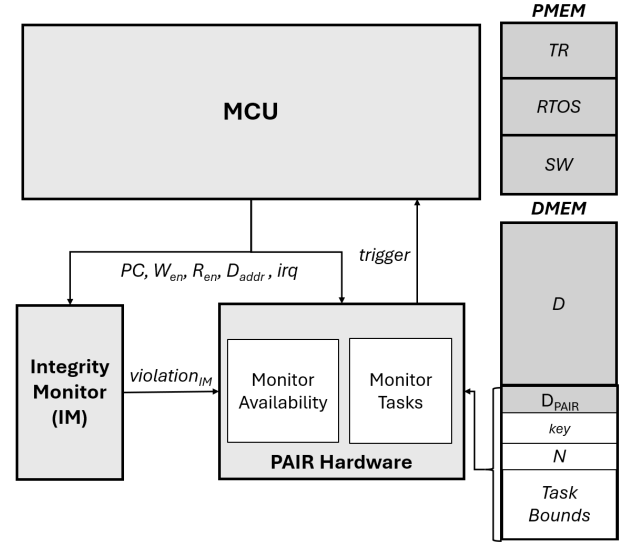


Figure 2: System Configuration: PAIR interfaces with MCU, an Integrity Monitor (IM), and memory

- W_{en} : a flag denoting whether the instruction is performing a write to memory;
- R_{en} : a flag denoting whether the instruction is performing a read from memory;
- D_{addr} : the memory address targeted by the current instruction (i.e., the address being written/read if W_{en}/R_{en});
- irq : MCU signal that indicates whether an interrupt is happening (i.e., the MCU has identified the interrupt source and is currently interrupting the current execution).

We assume the MCU is extended with an existing IM that interfaces with relevant MCU signals. We assume the IM outputs a signal $violation_{IM}$ that is only set when the current instruction is violating its security policy. This signal is passed into PAIR hardware.

MCU memory is divided into program memory (PMEM) and data memory (DMEM). PMEM contains fixed sub-regions for the following components:

- TR: the task region containing all real-time tasks;
- RTOS: the region containing the RTOS itself;
- SW: the region containing PAIR trusted software.

DMEM contains the following fixed sub-regions:

- D: memory used for tasks and RTOS;
- D_{PAIR} : all critical data referenced by SW or hardware.

D_{PAIR} contains a *key* used by SW trusted update function, the number of tasks configured in TR (N), and their bounds within TR (i.e., for each T_i , the task bounds are specified as a T_i^{min} and T_i^{max} holding the minimum and maximum PMEM addresses pertaining to T_i). We assume that task bounds are non-overlapping and independent in terms of their code, but may share data. PAIR hardware monitors signals from MCU and IM to monitor tasks and their availability (i.e., maintain AR), and outputs a *trigger*, which implements a non-maskable interrupt to invoke SW.

Definition 1: Post-Violation Availability

$$G : \{ \text{violation}_{\text{PAIR}} \vee \text{violation}_{\text{IM}} \rightarrow X(\text{trigger}) \}$$

$$G : \{ (PC \in TR) \wedge (task_{id} = i) \wedge \text{trigger} \rightarrow \neg X(T_i \in AR) \}$$

Definition 2: Post-Violation Integrity

$$G : \{ (PC \in TR) \wedge (task_{id} = i) \rightarrow (T_i \in AR) \vee X(\text{trigger}) \}$$

Figure 3: Formal Specification of security properties

4.3 Formalized Security Properties

We formalize the security properties of PAIR in Figure 3. We define [P1] as the two LTL statements in Definition 1. Given a run-time integrity violation is occurring ($\text{violation}_{\text{IM}}$ is true) or another access-control violation to PAIR critical components ($\text{violation}_{\text{PAIR}}$ is true), PAIR’s kill-and-yield trigger (trigger) should be set in the next state. Given task i (T_i) is executing (i.e., PC is within the bounds of TR and the current $task_{id}$ is i) and PAIR has generated trigger , the violating task T_i should no longer be in AR in the next state. We define [P2] in Definition 2. If T_i is executing, it must also be true that either T_i is in AR or that trigger will be set in the next state.

5 Security Sub-properties and Proofs

5.1 PAIR module Sub-properties

Based on PAIR’s goals, we outline a set of required sub-properties to be implemented by PAIR’s design. These sub-properties fall into two classes: task monitoring and availability monitoring. Later in Section 5.2, we demonstrate that these sub-properties suffice to achieve PAIR’s desired security properties.

An axiom as a part of the task monitoring specification is the interaction with D_{PAIR} to read the task boundaries. This is outlined in Figure 4. Given a fixed set of tasks N with their bounds installed in fixed locations, we assume LTL 1 that specifies the configuration of an internal label $task_{id}$. This label is set depending on which task bounds PC is currently within. Given PC is within the bounds of T_i , $task_{id}$ is set to i .

Two additional sub-properties are required to ensure that tasks do not perform any illegal memory accesses. These sub-properties are also outlined in Figure 4. The sub-property specified in LTL 2 requires that PAIR trigger SW to remove task i from AR when that task attempts to read or write to D_{PAIR} while outside of SW. This is defined by setting $\text{violation}_{\text{PAIR}}$ when a read or write instruction is targeting D_{PAIR} while not executing within SW. The second sub-property specifies that $\text{violation}_{\text{PAIR}}$ should be set when software that is not in SW attempts to modify PMEM, as defined in LTL 3. Whenever $\text{violation}_{\text{PAIR}}$ is set, trigger must be set in the next state, as specified in LTL 4.

Note: Further access control is possible if implemented in the IM. For example, our prototype IM (described in Section 7) enforces CFI and prevents tasks from modifying each other’s stacks.

The required sub-properties for monitoring availability and maintaining AR are outlined in Figure 5. PAIR maintains a bit vector AR_{en} that signifies whether a task is a part of the AR : removing or adding a task to AR corresponds to clearing or setting its corresponding bit in AR_{en} . Therefore, changes to AR are formally specified by bitwise operations that clear or set corresponding bits in AR_{en} . To model this, we define $\text{mask}(id)$ as a method to map an id to an N -length bitmask. Similarly, we define zero and set as N -length bitmasks of all zeros and ones, respectively.

LTL 5 and 6 specify the requirements to control PAIR’s non-maskable interrupt signal trigger . LTL 5 shows that trigger should be set in the next state when $\text{violation}_{\text{IM}}$ has been set by the IM. LTL 6 shows that trigger is cleared in the next state when trigger is currently set and the irq has been set, denoting that the interrupt has been accepted by the MCU.

LTL 7 requires that the bit in AR_{en} corresponding to the executing task is cleared when trigger has been generated. This is specified by requiring that the bitwise AND of $\text{mask}(task_{id})$ and AR_{en} in the next state equals zero.

Any attempts to re-execute a task that has been removed from AR should also result in the invocation of PAIR’s software to yield to another non-violating task. This is specified by LTL 8 by requiring that trigger is set in the next state when the bit in AR_{en} corresponding to T_i has been cleared. Finally, AR should only be reset (i.e., all bits in AR_{en} set to one) when exiting from SW. This is specified in LTL 9.

5.2 PAIR goals via sub-properties

Figure 6 outlines the theorems that state properties [P1] and [P2] are upheld. These theorems are also checked via model checker, described further in Section 7. Here we also discuss the intuition behind why they are upheld.

Theorem 1 states that LTLs 2, 3, 4, 5, and 6 uphold Definition 1 for [P1]. Definition 1 specifies that each instance of $\text{violation}_{\text{IM}}$ or $\text{violation}_{\text{PAIR}}$ must always result in trigger in the next state. It is provided by LTLs 5 and 6 that trigger is set at this instance and is not cleared until the MCU generates its interrupt signal, denoting it was accepted. Additionally, LTLs 2 and 3 ensure that a task’s memory accesses relevant to PAIR result in trigger , regardless of the specific IM in use.

Theorem 2 states that LTLs 1, 7, 8, 9 uphold Definition 2 for [P2]. Definition 2 describes post-violation integrity by specifying that execution of a task in a given state implies either that the task is in AR or that trigger will be generated in the next state. This property requires monitoring of the currently executing task, which is specified in LTL 1 to be maintained in $task_{id}$. LTL 7 specifies that given trigger has been generated, the current task is removed from AR by per-

Axiom: Tracked Task Execution:

$$G : \{PC \leq T_i^{max} \wedge PC \geq T_i^{min} \rightarrow (task_{id} = i)\} \forall i \in \{0, N-1\} \quad (1)$$

Task Memory Access Integrity:

$$G : \{(R_{en} \vee W_{en}) \wedge (D_{addr} \in D_{PAIR}) \wedge \neg(PC \in SW) \rightarrow violation_{PAIR}\} \quad (2)$$

$$G : \{W_{en} \wedge (D_{addr} \in PMEM) \wedge \neg(PC \in SW) \rightarrow violation_{PAIR}\} \quad (3)$$

$$G : (violation_{PAIR} \rightarrow X(trigger)) \quad (4)$$

Figure 4: Sub-properties to monitor task execution.

Definitions: N -bit bitmasks.

$$mask(id) := (1 \ll id), zero := 0_N, set := 1_N$$

Triggering upon integrity monitor violation:

$$G : \{violation_{IM} \rightarrow X(trigger)\} \quad (5)$$

$$G : \{trigger \wedge irq \rightarrow \neg X(trigger)\} \quad (6)$$

Reducing AR coverage upon trigger:

$$G : \{((PC \in TR) \wedge (task_{id} = i)) \wedge trigger \rightarrow (X(AR_{en} \ \& \ mask(task_{id})) = zero)\} \quad (7)$$

Triggering upon re-execution of violating task:

$$G : \{((PC \in TR) \wedge (task_{id} = i)) \wedge (AR_{en} \ \& \ mask(task_{id})) = zero \rightarrow X(trigger)\} \quad (8)$$

Resetting AR after software update:

$$G : \{PC = SW_{exit} \rightarrow X(AR_{en} = set)\} \quad (9)$$

Figure 5: Sub-properties to maintain AR.

Theorem 1:

LTLs 2, 3, 4, 5-6 $\rightarrow$ Definition 1

Theorem 2:

LTLs 1, 7, 8, 9 $\rightarrow$ Definition 2

Figure 6: Theorems that [P1] and [P2] are upheld.

forming the corresponding bitwise operations based on $task_{id}$. Additionally, LTL 8 supports Definition 2 by ensuring any attempted re-entering of a removed task results in *trigger* generation, specified by generating *trigger* when AR_{en} shows the current task was revoked from AR. Finally, LTL 9 also supports Definition 2 by specifying tasks can be re-added to AR after exiting from SW, and thus they will not generate *trigger* if they have been re-executed after a software update.

6 Implementation & Verification

PAIR modules are implemented as two finite state machines (FSMs) controlling *trigger* and AR_{en} , respectively. These are depicted in Figures 7 and 8.

6.1 Sub-module FSMs

Figure 7 shows the FSM for handling *trigger*. The value of *trigger* is set based on either of the following states:

- *Exec*: representing the state in which the MCU is executing without a violation, hence *trigger* is unset;
- *Trigger*: representing the state in which a violation has occurred and *trigger* is set.

The MCU enters *Exec* upon any reset, and thus this is the initial state. A transition from *Trigger* into *Exec* occurs based on the specification in LTL statements 2, 3, and 5 corresponding to run-time violations. A transition from *Trigger* into *Exec* only occurs when the *trigger*-based interrupt has been accepted by the MCU, corresponding to LTL statement 6.

Figure 8 shows the FSM for modifying AR via the bitvector AR_{en} . Each state corresponds to the status of AR_{en} :

- *Monitor*: state corresponding to execution that in which tasks are monitored. This state does not require reinstating or revoking a task to/from AR; thus, AR_{en} remains unchanged in this state;
- *Revoke*: state corresponding to that in which a task must be removed from AR, and the corresponding bit of the violating task is cleared in this state;
- *Reinstate*: state corresponds to a successful software update and thus all tasks are reinstated in AR.

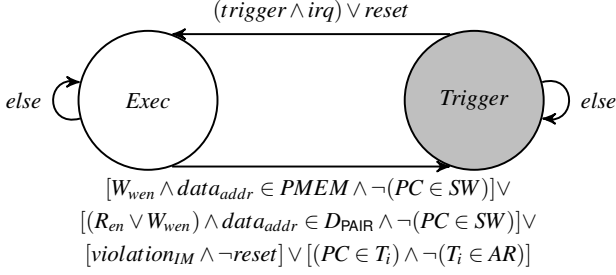


Figure 7: Verified FSM for LTLs 2, 3, 5, 6.

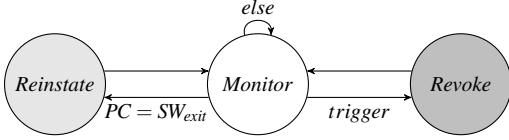


Figure 8: Verified FSM for LTLs 7, 9.

This FSM initializes in the *Monitor* state. Only upon a trigger does a transition into *Revoke* occur, and a transition from *Monitor* into *Reinststate* occur only occurs when a software update completes (i.e., PC being equal to SW_{exit}). Both *Revoke* and *Reinststate* states only execute for one cycle to update AR accordingly before transitioning to *Monitor*. The transition from *Monitor* into *Revoke* corresponds to LTL 7, and the transition from *Monitor* into *Reinststate* corresponds to LTL 9.

6.2 Formal Verification

We use NuSMV [17] as a model checker to determine whether PAIR design adheres to the specified properties and statements from Section 5 are upheld. We use Verilog2SMV [30] to convert Verilog specifications into nuSMV statements. Then, we formulate the LTL specifications in nuSMV to check the model. We execute the model checking atop an Intel(R) Core(TM) Ultra 7 155H (3.80 GHz) with 32.0 GB RAM using Ubuntu-20.04 under Windows Subsystem for Linux 2 (WSL 2). The model checker executed for 43.98 seconds with peak memory usage of 1.38GB to complete the verification.

7 Prototype & Evaluation

7.1 Prototype configuration

To implement PAIR prototype, we use the Xilinx Vivado tool-set [65] as a development platform. PAIR hardware is written in the Verilog hardware description language and implements each of PAIR sub-modules according to the logic defined in Section 6, and its software is implemented in C. Using Vivado, we synthesize PAIR hardware, simulate behavior, evaluate area/energy costs, and deploy a PAIR-equipped open-

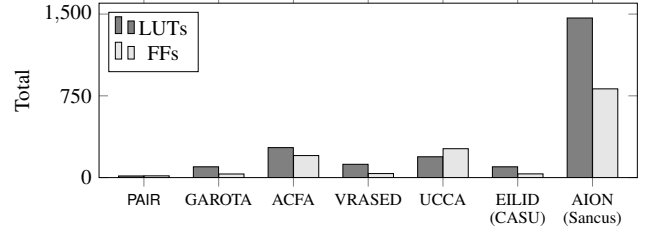


Figure 9: Hardware cost of PAIR is negligible and significantly lower than related works.

MSP430 [25] on the Basys3 prototyping board [22] with 16KB of PMEM and 12KB of DMEM.

A custom linker is used to ensure all code and data of interest (from Figure 2) are placed into contiguous memory regions that are monitored by PAIR. For the RTOS in our evaluation, we use RIOT [52] due to its open-source availability, support for MCUs within our scope, and use in prior works [2]. In theory, PAIR concepts can apply to any RTOS with an identifiable API for *kill* and *yield* procedures.

For the IM, we instantiate a task-aware shadow stack module along with an indirect-branch table (IBT) in hardware for CFI of return and jump instructions, respectively. Both interface with memory and MCU signals as required. Additionally, it monitors tasks' stacks to ensure they do not interfere with each other's temporary variables.

7.2 Fixed Costs

PAIR hardware cost is accounted for in the Look-up tables (LUTs) and flip-flop registers (FFs) added to the openMSP430 core. We compare this cost to that of closely related works in Figure 9. PAIR requires fewer resources than all closely related works that deploy security extensions atop openMSP430. Although PAIR has a different goal than each of these works, this shows that it is cost-effective for low-end devices and is in line with related works that target the same class of devices. The most closely related works are EILID [33] and AION [2]. EILID implements CFI for low-end MCUs atop a hybrid (software-hardware) root of trust CASU [21] for memory immutability and trusted software updates. We note that when including IM in the cost, PAIR + IM requires more hardware resources than EILID. However, we note that PAIR is concerned with providing availability and run-time integrity simultaneously, whereas EILID's primary focus is run-time integrity (discussed further in Section 10). AION builds atop Sancus [46] hardware-based TEE, which requires a more extensive set of security features. We also evaluate PAIR performance overheads by considering additional energy consumption. To evaluate energy overheads, we use Vivado synthesis tools to estimate PAIR's power consumption on the Basys3 FPGA board. We find the openMSP430 core without PAIR consumes 71 mW of static power and 76 mW with PAIR, showing that PAIR incurs only an additional 5 mW in static

Table 1: Memory usage for applications with multiple tasks (MT), recursion (R), indirect forward branches (IF), and high-/very high MCU suitability (SUI) [51].

Application	Attribute	Usage in bytes	
		PAIR	PAIR + IM
sched_round_robin	MT	12 (+0.073%)	24 (+0.146%)
ipc_pingpong	MT	12 (+0.073%)	14 (+0.085%)
thread_priority_inversion	MT	16 (+0.097%)	22 (+0.134%)
mutex_sleep	MT	8 (+0.048%)	12 (+0.073%)
sched_change_priority	MT	8 (+0.048%)	12 (+0.073%)
crc_32	SUI	4 (+0.024%)	8 (+0.048%)
matrixmult	SUI	4 (+0.024%)	12 (+0.073%)
lcdnum	IF, SUI	4 (+0.024%)	42 (+0.256%)
cover	IF, SUI	4 (+0.024%)	374 (+2.28%)
cubic	SUI	4 (+0.024%)	10 (+0.061%)
fdct	SUI	4 (+0.024%)	10 (+0.061%)
tarai	R	4 (+0.024%)	18 (+0.109%)
recursion	R	4 (+0.024%)	26 (+0.159%)

power consumption. The dynamic power consumption of the simulated MCU board was measured at 109 mW. The contribution from PAIR was negligible, failing below Vivado’s minimum reportable threshold of 1 mW.

7.3 Performance costs

We evaluate the performance costs of PAIR using the run-time and memory overheads when executing software atop PAIR. PAIR hardware does not impose any run-time overheads, since it is monitoring MCU signals without affecting the MCU’s critical path. The memory cost incurred by PAIR varies per application. For instance, D_{PAIR} size depends on the number of independent tasks. Memory cost may also vary based on the particular IM in use. Our IM instance memory usage for shadow stacks depends on the maximum call depth among all tasks, and IBT size depends on the total number of indirect forward edge control flow transfers among all tasks. To effectively evaluate PAIR memory usage, we select a set of applications from RIOT test examples and BEEBS embedded systems benchmark suite [51]. We select RIOT test examples that create multiple tasks, and BEEBS applications that meet one or more of the following criteria:

1. has one or more recursive calls;
2. has one or more indirect forward branches (jump/call);
3. has *high* or *very high* suitability for MCUs [51].

Table 1 shows the resulting memory usage.

Among the selected applications, PAIR memory usage is minimal, an increase ranging from just 4 to 16 bytes (0.024% to 0.097% of available DMEM). Additional memory usage is required due to the IM, but it is still minimal: an increase ranging from 8 to 42 bytes (0.048% to 2.28% of the available data memory). PAIR by itself has a minimal fixed memory cost ($2 + 2 \times N$ bytes), but incurs additional variable costs due to the IM that is installed.

Finally, we evaluate the latency overhead introduced by PAIR’s interrupt path relative to the native software call of the *kill-and-yield* routine. We measure this latency across four

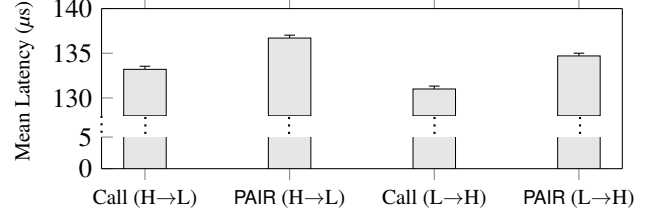


Figure 10: Mean *kill-and-yield* latency of PAIR interrupt by priority direction (high-to-low: H→L, vs. low-to-high: L→H) compared to a native call

scenarios, varying two factors. First, we vary the direction of the priority switch from high-to-low (H→L) and from low-to-high (L→H). Second, we vary the invocation source between the native call vs. PAIR-triggered interrupt, the handler of which jumps to the same function.

As shown in Figure 10, PAIR interrupt path introduces a consistent overhead of approximately $3.5\mu\text{s}$ relative to the native call path across both priority directions ($136.7\mu\text{s}$ vs. $133.2\mu\text{s}$ for H→L and $134.7\mu\text{s}$ vs. $131.0\mu\text{s}$ for L→H), reflecting the cost of hardware violation detection and non-maskable interrupt generation. The H→L transition incurs marginally higher latency in both cases, due the scheduler identifying the next task. The negligible and consistent times confirm that PAIR introduces minimal overhead.

8 Response Time Analysis (RTA)

Terminating a violating task frees system execution time, reducing the response time of remaining lower-priority tasks, but this benefit may be offset by PAIR’s interrupt-path latency (see Section 7.3). To analyze this tension, we apply fixed-priority RTA: we first establish a baseline RTA model, extend it to capture PAIR’s revocation behavior and measured interrupt-path cost, then derive the condition under which interference from a higher-priority task eliminated by revocation offsets this enforcement cost. Table 2 summarizes notation.

8.1 Baseline RTA Model

We first establish a baseline using classical fixed-priority RTA [34]. For a task i , the worst-case response time is given by the least non-negative fixed point of Equation 10:

$$R_i^{(n)} = C_i + B_i + \sum_{j \in hp(i)} \left\lceil \frac{R_i^{(n-1)}}{T_j} \right\rceil C_j, \quad (10)$$

In this equation, C_i and B_i denote the worst-case execution time and blocking time of task i , respectively. The terms C_j and T_j denote the worst-case execution time and period of higher priority task j . The set $hp(i)$ denotes all tasks with higher priority than task i . For each task j in $hp(i)$, the ceiling term $\lceil R_i^{(n-1)} / T_j \rceil$ gives the number of times task j can

Table 2: Response Time Analysis Notation

Notation	Meaning
C_i, C_k, C_l	the WCET of tasks i, k , and l , respectively
B_i	the blocking time of task i
$hp(i)$	the set of tasks that are higher priority than task i
T_i, T_k, T_l	the period of tasks i, k , and l , respectively
$R_i^{(n)}$	the worst-case response time of task i computed at iteration n
R_i	the worst-case response time of task i after convergence
$budget_k$	the amount of execution time consumed by task k before PAIR revokes it
R_i^{baseline}	the converged worst-case response time of task i in the baseline model
n_k, n_l	simplified ceiling terms of task k and l in the baseline model (i.e., the number of times tasks k and l execute during the response time of task i in the baseline model)
$R_i^{\text{PAIR},0}$	the converged worst-case response time of task i in the post-violation state model without PAIR interrupt latency.
R_i^{PAIR}	the converged worst-case response time of task i in the post-violation state model with PAIR interrupt latency.
C_{PAIR}	the interrupt path latency introduced by PAIR, measured in Section 7.3
n_p	simplified ceiling term of task l in the post-violation model, shared by $R_i^{\text{PAIR},0}$ and R_i^{PAIR} (i.e., the number of times task l executes within the response time of task i , given task k only executes for $budget_k$ time)

execute within task i 's response time interval as estimated at the previous iteration. It is multiplied by C_j to give the task's total interference contribution. The interference contributions of all higher priority tasks are summed to obtain the total interference, which is summed with C_i and B_i to obtain the worst-case response time.

Because R_i appears on both sides of the equation, this is a recursive (fixed-point) equation rather than a closed-form expression. The notation $R_i^{(n)}$ denotes R_i at iteration n . Starting from $R_i^{(0)} = C_i + B_i$, the equation is iterated until it converges (i.e., when $R_i^{(n)} = R_i^{(n-1)}$) or the response time is deemed to exceed task i 's deadline. Upon convergence, $R_i^{(n)}$ becomes the fixed-point R_i . For all remaining equations, we assume R_i has already converged and thus simplify $R_i^{(n)} = R_i^{(n-1)} = R_i$.

For the baseline analysis, we consider three benign tasks: the task under analysis i and two higher-priority tasks, k and l , both of which are assumed to execute according to their normal periodic behavior. The baseline response time of task i in this scenario is consequently expressed in Equation 11:

$$R_i^{\text{baseline}} = C_i + B_i + \left\lceil \frac{R_i^{\text{baseline}}}{T_k} \right\rceil C_k + \left\lceil \frac{R_i^{\text{baseline}}}{T_l} \right\rceil C_l \quad (11)$$

In the baseline model, task k is not exploited and runs as intended, providing a reference point for the next scenario. In

the post-violation model next, task k models the exploited task, which PAIR detects and removes. We simplify Equation 11 using constants n_k and n_l to represent the ceiling of tasks k and l , obtaining the baseline worst-case response time (R_i^{baseline}) of task i in Equation 12:

$$R_i^{\text{baseline}} = C_i + B_i + n_k C_k + n_l C_l \quad (12)$$

8.2 Post-Violation State Model

We next model the system state after PAIR has removed a violating task. In particular, task k is assumed to execute for some bounded amount of time before a violation was detected, at which point it will be removed and prevented from any further execution by PAIR. Let $budget_k$ denote the amount of execution consumed by task k before it was removed from AR by PAIR. Since task k can execute for any portion of its expected execution time, the model permits $budget_k$ to hold any value between 0 and $n_k C_k$.

Following revocation, task k contributes no further periodic interference to the response-time of task i . Its contribution is therefore represented as a one-shot term, yielding the response-time equation in Equation 13:

$$R_i^{\text{PAIR},0} = C_i + B_i + budget_k + \left\lceil \frac{R_i^{\text{PAIR},0}}{T_l} \right\rceil C_l \quad (13)$$

In Equation 13, the superscript PAIR,0 denotes the idealized PAIR case with zero enforcement overhead. This formulation preserves the execution interference that has already occurred before revocation while preventing future execution of k . However, PAIR's interrupt path introduces a measurable delay (recall from Section 7.3) between the detection of a violation in task k and the yield into a pending task. This behavior must be incorporated into the response-time model to determine whether the non-degradation property holds for the implemented system. To account for this, we extend the response-time equation with a constant interrupt path latency, denoted by C_{PAIR} . The resulting response-time equation is shown in Equation 14:

$$R_i^{\text{PAIR}} = C_i + B_i + budget_k + C_{\text{PAIR}} + \left\lceil \frac{R_i^{\text{PAIR}}}{T_l} \right\rceil C_l \quad (14)$$

This model captures both competing effects of PAIR: the removal of future interference from k and the finite cost required to perform the *kill-and-yield* operation.

The value of C_{PAIR} is obtained from measurements of the PAIR prototype in Section 7.3: between 134.7 μ s and 136.7 μ s. Accordingly, we update the model to set C_{PAIR} equal to the rounded-up integer value of their mean: 136 μ s. Additionally, we simplify the expression for R_i^{PAIR} by using n_p to represent the ceiling factor in the post-violation model (e.g., the number of times task l interferes with task i), obtaining the final expression in Equation 15:

$$R_i^{\text{PAIR}} = C_i + B_i + \text{budget}_k + n_p C_l + 136 \quad (15)$$

8.3 Solving for Degradation Condition

We use the final simplified Equations 12 and 15 to represent the response time of task i under the two described scenarios. We perform symbolic analysis to determine the condition under which the response time of task i executing on an PAIR-enabled system degrades compared to the baseline system. We implemented this using SymPy [58] in Python using 120 lines of code.

The variables C_i , B_i , n_k , C_k , C_l , n_l , budget_k , and n_p are defined as symbolic variables, while C_{PAIR} is initialized to concrete integer value 136. R_i^{baseline} from Equation 12 and R_i^{PAIR} from Equation 15 are defined as expressions using the corresponding symbolic variables.

Our solver implemented using SymPy takes the difference of R_i^{PAIR} and R_i^{baseline} to generate a new expression corresponding to the non-degradation condition: $R_i^{\text{PAIR}} - R_i^{\text{baseline}} = 0$. The evaluation of this expression returned the non-degradation condition in Equation 16:

$$n_k C_k \geq \text{budget}_k + 136 + C_l(n_p - n_l). \quad (16)$$

This expression states that PAIR's interruption and removal of task k does not degrade task i 's response time as long as task k 's baseline interference (e.g., its execution count n_k multiplied by its execution time C_k) is at least as large as the combined cost of intervention:

1. the execution that task k completed before being interrupted (budget_k);
2. PAIR's interrupt path latency ($C_{\text{PAIR}} \approx 136 \mu\text{s}$);
3. the added interference from task l (the term $C_l(n_p - n_l)$) caused by task l executing a different number of times once k stops competing for processor time.

When PAIR does not affect the number of task l executions (e.g., $n_p = n_l$), this constraint can simplify into the more intuitive expression shown in Equation 17:

$$n_k C_k - \text{budget}_k \geq 136. \quad (17)$$

This scenario would occur when task l has higher priority than task k , while task k has higher priority than task i : since task l already preempts task k regardless of whether k is revoked, removing k does not change how often l executes.

In this case, degradation is determined purely by the budget_k value. To analyze this case further, we assume that budget_k takes the form $n_{bv} C_k$, where n_{bv} represents the number of complete executions of task k that occur *before a violation*. We use this substitution to obtain Equation 18:

$$(n_k - n_{bv}) C_k \geq 136. \quad (18)$$

This states a direct relationship between PAIR's enforcement cost and the amount of execution eliminated by revoking

task k . When $n_{bv} = n_k - \frac{136}{C_k}$, enforcement by PAIR does not impact the response-time of task i . A lower n_{bv} (i.e., violation by task k detected earlier) will result in an improved response time for task i . A higher n_{bv} (violation detected later) will incur an overhead bounded by the enforcement time duration. Note that any violation by a lower priority task has no effect on the response time of i .

9 Case Study

As a case study, we simulate the application example from Section 3.1, showing that PAIR enables a system with a compromised task to remain online until other tasks complete execution. The simulated system includes four tasks:

- Task 1: A syringe pump motor and dosage control [61];
- Task 2: A sensor to monitor a patient's temperature [55];
- Task 3: A communication module using the open-MSP430's peripheral UART interface;
- Task 4: A liquid crystal display [6].

Task 1 implements a stepper motor that advances the syringe plunger in micro-steps, with each pulse corresponding to one step. Based on the syringe dimensions and threaded rod pitch, the program computes approximately 13 micro-steps per milliliter (ml), yielding approximately 65 motor steps for a fixed 5 ml bolus. We modify this implementation with a counter, initialized to zero, that updates GPIO output port `p3_out` with the latest pulse count; a functional execution therefore ends with `p3_out` equal to 64 (0x40).

We evaluate this end-to-end example in two scenarios: (1) a benign execution and (2) an execution in which the communication module (task 3) is attacked via a simulated buffer overflow in input parsing, exploited to hijack the program's control flow. Waveforms for both appear in Figures 11 and 12.

Both figures share four key signals. `p3_out` is a GPIO peripheral output showing task 1's pulse-step counter. The remaining three signals capture PAIR's run-time behavior. `AR_en` is the bitmask representation of AR stored internally to PAIR hardware. In this case study, PAIR supports 16 concurrent tasks, so `AR_en` is a 16-bit bitmask initialized to 0xffff since all tasks initially belong to AR. `cfi_violation` is an input signal from IM indicating a detected CFI violation. Finally, `zombify_irq` is PAIR's non-maskable interrupt output, which triggers the RTOS's *kill-and-yield* functionality and simultaneously updates `AR_en` when set.

The remaining waveform signals are monitoring signals added during simulation to identify which module is executing. `in_task1`, `in_task2`, `in_task3`, and `in_task4` are binary signals indicating when tasks 1 through 4 are executing, respectively, computed by comparing `PC` to each task's static memory bounds in program memory (omitted from the waveforms for brevity). Similarly, `PC` is compared to the RTOS region's bounds to produce `in_RTOS`, signaling when the RTOS is executing.

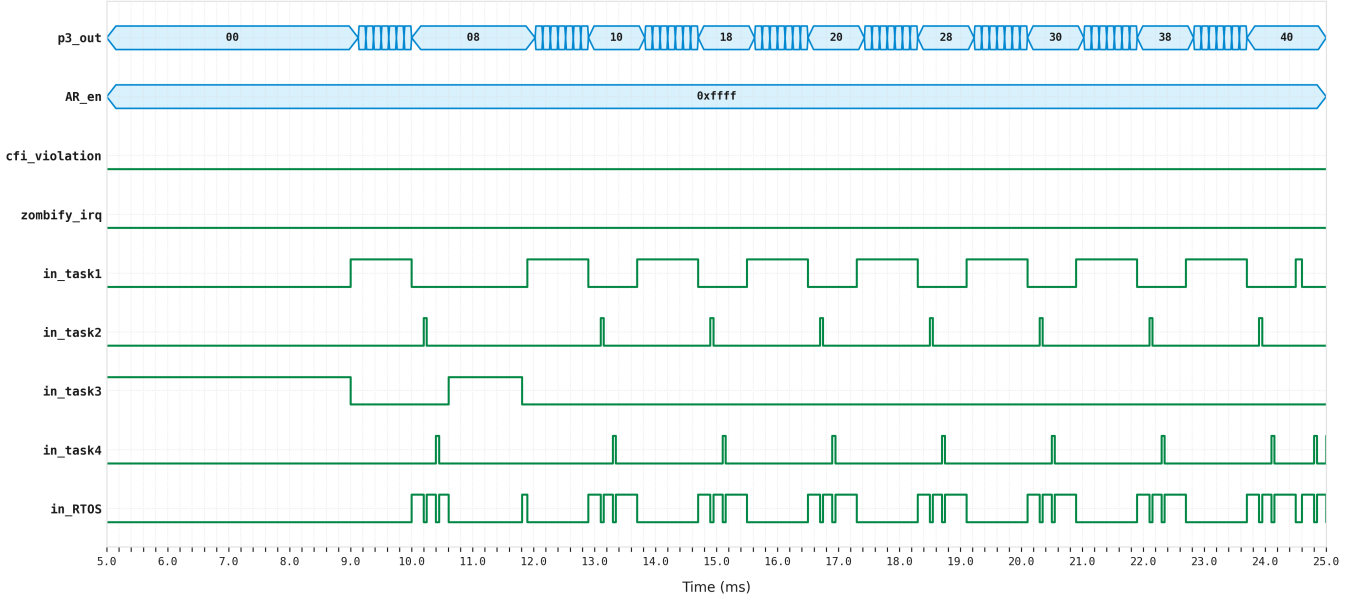


Figure 11: Case study: this waveform shows the result of a benign execution. During this execution, no `cfi_violation` occurs, and thus PAIR does not trigger its `zombify_irq` signal. As a result, the syringe pump control task (task_1) outputs its dosage in 64 steps (shown through its output on `p3_out` showing `0x40`). All tasks complete by ≈ 26.45 ms, with the syringe pump control completing by ≈ 24.62 ms.

Figure 11 shows the waveform for the benign execution. Task 3 runs first, receiving input from the network. Once a command is parsed, the RTOS schedules the remaining tasks, which then execute periodically with intermittent RTOS switching between them. No CFI violation is detected during this execution, so IM never sets `cfi_violation`, PAIR’s `zombify_irq` remains cleared throughout, and `AR_en` stays at its initialized value `0xffff`, indicating no task was removed from AR. The syringe control task completes at ≈ 24.62 ms, having executed all dosage pulses, as reflected by `p3_out` reaching `0x40` at the end of its execution. The full system completes at ≈ 26.45 ms.

Figure 12 showcases the waveform after simulating the attacked execution. In this scenario, a CFI violation is detected during the execution of the communication module (task 3). This occurs at ≈ 10.71 ms, as shown with the rising edge of `cfi_violation` at this time. This corresponds to PAIR setting `zombify_irq` to trigger the removal of task 3 via the RTOS’s *kill-and-yield* functionality. At this same time, PAIR updates AR to remove the violating task. This is shown in the transition of `AR_en` from `0xffff` to `0xffffb`.

At this point, task 3 is no longer authorized to execute. Compared to the benign execution in Figure 11, the second pulse of `in_task3` is much shorter, and it does not continue after the `cfi_violation` is set. This portion of the waveform shows that task 3 is prevented from further execution. However, similarly to the benign case shown in Figure 11, the RTOS follows the same schedule and enters back into task 1,

shown with `in_task1` being the next task monitor signal to be set after the second pulse of `in_task3`.

The three remaining non-violating tasks (task 1, task 2, and task 4) continue executing until all have completed their operation. In this scenario, due to the removal of task 3 midway through its execution, the entire system completes execution early. In this case, all tasks finish executing at ≈ 25.31 ms compared to ≈ 26.45 ms in the benign execution (≈ 1.14 ms early). Similarly, task 1 finishes early compared to the benign execution: in this scenario, it completes at ≈ 23.47 ms compared to ≈ 24.62 ms in the previous execution.

This case study demonstrates PAIR’s non-degradation property in a realistic, multi-task embedded system rather than just analytically. When task 3 is compromised and revoked mid-execution, the remaining tasks are unaffected in their functional correctness, such as the syringe pump control task completing all of its expected pulses. Rather than degrading system performance, revocation actually allows the system to finish earlier since the compromised task no longer competes for CPU time. This confirms that PAIR can remove a violating task at run time without disrupting the timing or correctness of the tasks that continue running.

10 Related Work

CFI in Real-Time Systems. Many works have explored CFI in embedded and real-time systems [42]. InsectACIDE [63] proposes a method for CFI for ARM Cortex-M MCUs lever-

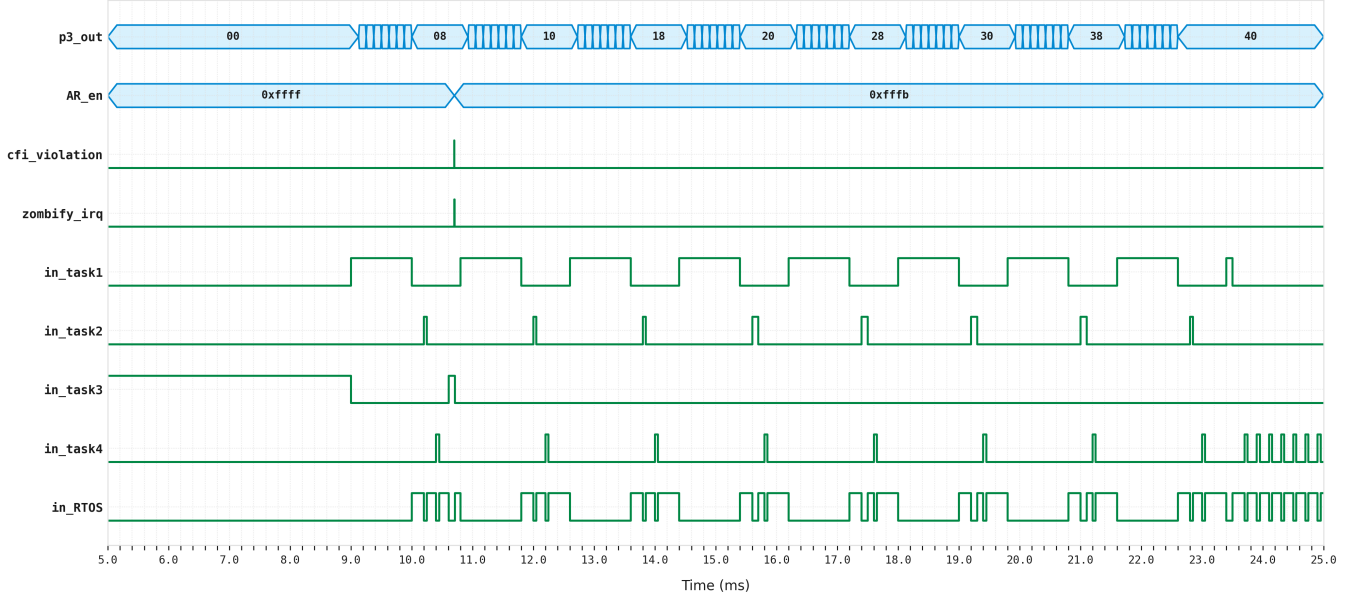


Figure 12: Case Study: this waveform shows the end result of an attacked execution, in which Task 3 (the communication module) has a CFI violation. At ≈ 10.71 ms, the `cfi_violation` signal is raised during the execution of Task 3. This results in the removal of Task 3 from AR (shown by transition of `AR_en=0xffff` to `0xfffb`) and a trigger of `zombify_irq`, PAIR’s non-maskable interrupt. By the end of the execution, all remaining tasks have completed in 25.31 ms, and the syringe dosage control completes at ≈ 23.47 ms with all dosages given (e.g., with `p3_out=0x40`, the expected value from the benign execution).

aging CoreSight tracing extensions to record control flow transfers without incurring run-time overhead. Then, recorded control flow transfers are checked for integrity violations during idle times. RECFISH [62] demonstrates an RTOS-compatible CFI method for ARM Cortex-R MCUs through shadow stacks and function-label checks via software instrumentation. FastCFI [23] also uses ARM extensions, but instead uses an FPGA to store and check a CFG. Regardless of their specificities, PAIR differs from each in that it accounts for the post-violation phase, allowing non-violating tasks to continue their execution.

EILID [33] proposes a hybrid method for low-end MCUs and evaluates atop openMSP430. It uses code instrumentation for trampolines that jump to a hardware-protected CFI monitor. PAIR +IM incurs more hardware overhead, while EILID incurs more run-time overhead and an increase in code size. Similar to PAIR, EILID triggers an interrupt upon violations and also enables software updates. However, EILID’s interrupt jumps directly to the software update function. PAIR’s interrupt instead disables a violating task before returning to the next high-priority benign task. Due to parallel goals, PAIR and EILID could coexist, using PAIR to provide post-violation availability/integrity with EILID as a low-cost IM.

Availability-focused mechanisms. Active roots of trust (ARoTs) [2, 3, 12, 14, 27, 38] ensure a trusted function is guaranteed to execute upon a corresponding hardware signal or upon MCU reset. The concept of ARoTs was proposed in

GAROTA [3], a formally verified architecture that provides two properties regarding the execution of a trusted interrupt service routine (ISR): (1) the guaranteed triggering of the trusted ISR upon its interrupt source (e.g., GPIO, timer), and (2) the guaranteed re-triggering of the trusted ISR upon boot or reset. Several works built upon GAROTA to realize ARoTs for various purposes [12, 14, 38].

AION [2] is an ARoT that builds upon Sancus TEE [46] to provide strong availability guarantees to a trusted scheduler in the presence of compromised software. Like AION, we also built our prototype using openMSP430 as an MCU and RIOT as an RTOS. AION ensures the availability of a trusted Sancus-aware scheduler by incorporating hardware-based atomicity monitor and exception engine modules. While both AION and PAIR invoke the RTOS scheduler upon a violation signal, they pursue complementary objectives. AION aims to preserve the availability of a TEE-protected trusted scheduler to fairly allocate execution time to tasks, whereas PAIR focuses on preventing tasks that violate run-time integrity policies from executing until they are updated, thereby preserving availability to other non-violating tasks on the same platform. This is provided even if other software, including the scheduler, attempts to re-enter it. Given the orthogonal goals, PAIR and AION combined could provide a strong security service: AION’s trusted scheduler with PAIR’s enforcement against re-entering of violating tasks could return both fair scheduling and post-violation containment.

SDRaD [27] leverages Intel PKU to compartmentalize applications and to trigger rollback of execution state upon a violation. SDRaD is tailored towards restoring prior execution context with custom handlers to avoid replay of the attack and targets rich application software on COTS processors, whereas PAIR targets resource-constrained MCUs operating under hard real-time constraints, enforcing post-violation task containment entirely in hardware, and explicit integration with RTOS scheduling semantics.

Formally verified MCU mechanisms. Several prior works share this work’s strategy of formal modeling/verification via LTL. VRASED [47] is a formally verified hybrid remote attestation architecture, using formally verified cryptographic libraries to compute attestation evidence and LTL to model and verify the hardware extension that monitors and protects the attestation software. APEX [48] and ASAP [13] propose formally verified architectures for *proofs of execution*, demonstrating that a given software executed completely and without modification, and that reported outputs were generated by that same execution. UCCA [59] is a formally verified architecture providing isolation of untrusted code, with formal verification ensuring its hardware extension provide return integrity, stack integrity, and critical memory region protection.

11 Discussion & Future Work

Integrity Monitor Soundness. PAIR’s guarantees are conditioned on the correctness of the IM, as PAIR takes $violation_{IM}$ as input from IM. We do not verify the correctness of the IM in use. A false negative in the IM leaves PAIR with nothing to act on, silently forfeiting desired properties despite PAIR behaving correctly. Conversely, a false positive would revoke a benign task, directly working against PAIR’s availability goal. As a result, PAIR’s security is only as strong as the soundness of the paired IM. As there have been proposals for formally-verified CFI mechanisms [15, 19, 37], pairing PAIR with such an IM would ensure end-to-end formal guarantees.

Interdependent Tasks. PAIR assumes tasks are functionally independent. It is possible that an application is designed such that its tasks coordinate via locks, mutexes, or semaphores. As such, removing a violating task from AR mid-execution may leave shared locks unreleased, causing deadlock. Addressing this requires further consideration of RTOS locking primitives, including monitoring of its access and ability to unlock in special cases from PAIR’s SW protected region. We leave this as a direction for future work.

Side-channel attacks. PAIR’s software update routine authenticates update requests using a secret key. A naive implementation may be vulnerable to timing side-channel attacks if it enables any secret-dependent execution paths. This can be mitigated by ensuring the authentication routine is implemented as constant-time code. Cache-timing side-channels are a common concern for cryptographic routines on general-purpose processors. However, this is not applicable to devices

in PAIR’s scope, which often do not incorporate caches. However, this is an important consideration when extending to more complex processors in future work.

Multi-Core Embedded Systems. Extending PAIR to multi-core embedded systems introduces challenges. For example, AR state must be maintained atomically across concurrent contexts, and kill-and-yield semantics must account for cross-core scheduling. An extension of PAIR to support multi-core is an intriguing direction for future work.

Task Criticality. PAIR provides an effective balance for scenarios in which preventing a task from continuing is safer than allowing it to continue in an unexpected state, regardless of the task’s criticality. Although applicable to many scenarios (as described in Section 3.1), there are scenarios in which preventing a compromised high-criticality task from continuing may still be unideal (e.g., automotive vehicles). Future work could explore enhancing PAIR SW to differentiate violation handling based on criticality: lower-criticality tasks are immediately removed from AR, while higher-criticality tasks are re-executed once before removal, leveraging aspects of PAIR and related works [2, 3, 27].

12 Conclusion

In this work, we propose PAIR to preserve task availability and system-wide run-time integrity. By ensuring that only the offending task is prevented from executing, PAIR enables non-offending critical real-time tasks to continue executing. We designed, implemented, and evaluated a formally verified and open-source prototype of PAIR¹, finding that its design has reasonable costs, incurring no runtime overhead while imposing minimal hardware and performance overheads.

References

- [1] Ali Abbasi, Thorsten Holz, Emmanuele Zambon, and Sandro Etalle. Ecfi: Asynchronous control flow integrity for programmable logic controllers. In *Proceedings of the 33rd Annual Computer Security Applications Conference*, pages 437–448, 2017.
- [2] Fritz Alder, Jo Van Bulck, Frank Piessens, and Jan Tobias Mühlberg. Aion: Enabling open systems through strong availability guarantees for enclaves. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*, pages 1357–1372, 2021.
- [3] Esmerald Aliaj, Ivan De Oliveira Nunes, and Gene Tsudik. GAROTA: generalized active Root-Of-Trust architecture (for tiny embedded devices). In *31st USENIX Security Symposium (USENIX Security 22)*, pages 2243–2260, 2022.

¹The source code will be made available after peer review.

- [4] Naif Saleh Almakhdhub, Abraham A Clements, Saurabh Bagchi, and Mathias Payer. μ RAI: Securing Embedded Systems with Return Address Integrity. In *Network and Distributed Systems Security (NDSS) Symposium*, 2020.
- [5] Mahmoud Ammar, Adam Caulfield, and Ivan De Oliveira Nunes. Sok: Integrity, attestation, and auditing of program execution. In *2025 IEEE Symposium on Security and Privacy (SP)*, pages 77–77. IEEE Computer Society, 2024.
- [6] Arduino. Liquidcrystal library for arduino, 2009.
- [7] ARM. Return Address Signing using ARM Pointer Authentication. <https://gcc.gnu.org/legacy-ml/gcc-patches/2018-11/msg00104.html>, 2018. [Online; accessed 13-February-2023].
- [8] Arm Ltd. Arm architecture reference manual for a-profile architecture. <https://developer.arm.com/documentation/ddi0487/1b>, 2025. Section C6.2.49 & D8.10.
- [9] Tyler Bletsch, Xuxian Jiang, Vince W Freeh, and Zhenkai Liang. Jump-oriented programming: a new class of code-reuse attack. In *Proceedings of the 6th ACM symposium on information, computer and communications security*, pages 30–40, 2011.
- [10] Nathan Burow, Scott A Carr, Joseph Nash, Per Larsen, Michael Franz, Stefan Brunthaler, and Mathias Payer. Control-flow integrity: Precision, security, and performance. *ACM Computing Surveys (CSUR)*, 50(1):1–33, 2017.
- [11] Nathan Burow, Xinping Zhang, and Mathias Payer. Sok: Shining light on shadow stacks. In *2019 IEEE Symposium on Security and Privacy (SP)*, pages 985–999. IEEE, 2019.
- [12] Adam Caulfield, Antonio Joia Neto, Norrathep Rattanavipanon, and Ivan De Oliveira Nunes. Traces: Tee-based runtime auditing for commodity embedded systems. *arXiv preprint arXiv:2409.19125*, 2024.
- [13] Adam Caulfield, Norrathep Rattanavipanon, and Ivan De Oliveira Nunes. ASAP: reconciling asynchronous real-time operations and proofs of execution in simple embedded systems. In *Proceedings of the 59th ACM/IEEE Design Automation Conference*, pages 721–726, 2022.
- [14] Adam Caulfield, Norrathep Rattanavipanon, and Ivan De Oliveira Nunes. ACFA: Secure runtime auditing & guaranteed device healing via active control flow attestation. In *32nd USENIX Security Symposium (USENIX Security 23)*, pages 5827–5844, 2023.
- [15] Katharina Ceesay-Seitz, Flavien Solt, and Kaveh Razavi. μ cfi: Formal verification of microarchitectural control-flow integrity. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, pages 213–227, 2024.
- [16] Nick Christoulakis, George Christou, Elias Athanassopoulos, and Sotiris Ioannidis. Hcfi: Hardware-enforced control-flow integrity. In *Proceedings of the Sixth ACM Conference on Data and Application Security and Privacy*, pages 38–49, 2016.
- [17] Alessandro Cimatti et al. Nusmv 2: An opensource tool for symbolic model checking. In *International conference on computer aided verification*, pages 359–364. Springer, 2002.
- [18] Crispin Cowan, F Wagle, Calton Pu, Steve Beattie, and Jonathan Walpole. Buffer overflows: Attacks and defenses for the vulnerability of the decade. In *Proceedings DARPA Information Survivability Conference and Exposition. DISCEX’00*, volume 2, pages 119–129. IEEE, 2000.
- [19] John Criswell, Nathan Dautenhahn, and Vikram Adve. Kcofi: Complete control-flow integrity for commodity operating system kernels. In *2014 IEEE symposium on security and privacy*, pages 292–307. IEEE, 2014.
- [20] Lucas Davi, Matthias Hanreich, Debayan Paul, Ahmad-Reza Sadeghi, Patrick Koeberl, Dean Sullivan, Orlando Arias, and Yier Jin. Hafix: Hardware-assisted flow integrity extension. In *Proceedings of the 52nd Annual Design Automation Conference*, pages 1–6, 2015.
- [21] Ivan De Oliveira Nunes, Sashidhar Jakkamsetti, Youngil Kim, and Gene Tsudik. CASU: Compromise avoidance via secure update for low-end embedded systems. In *Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design*, pages 1–9, 2022.
- [22] Digilent. Basys 3 artix-7 FPGA trainer board: Recommended for introductory users, 2018.
- [23] Lang Feng, Jeff Huang, Jiang Hu, and Abhijith Reddy. Fastcfi: Real-time control-flow integrity using fpga without code instrumentation. *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 26(5):1–39, 2021.
- [24] Alexander J Gaidis, Joao Moreira, Ke Sun, Alyssa Milburn, Vaggelis Atlidakis, and Vasileios P Kemerlis. Fineibt: Fine-grain control-flow enforcement with indirect branch tracking. *arXiv preprint arXiv:2303.16353*, 2023.
- [25] Olivier Girard. openMSP430, 2009.

- [26] Michele Grisafi, Mahmoud Ammar, Marco Roveri, and Bruno Crispo. Flashadow: A flash-based shadow stack for low-end embedded systems. *ACM Transactions on Internet of Things*, 5(3):1–29, 2024.
- [27] Merve Gülmez, Thomas Nyman, Christoph Baumann, and Jan Tobias Mühlberg. Rewind & discard: Improving software resilience using isolated domains. In *2023 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, pages 402–416. IEEE, 2023.
- [28] Hong Hu, Chenxiong Qian, Carter Yagemann, Simon Pak Ho Chung, William R Harris, Taesoo Kim, and Wenke Lee. Enforcing unique code target property for control-flow integrity. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*, pages 1470–1486, 2018.
- [29] Texas Instruments. Application report slaa685: Msp code protection features. <https://www.ti.com/lit/an/slaa685/slaa685.pdf>, 2015.
- [30] Ahmed Irfan, Alessandro Cimatti, Alberto Griggio, Marco Roveri, and Roberto Sebastiani. Verilog2smv: A tool for word-level verification. In *2016 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pages 1156–1159. IEEE, 2016.
- [31] Md Rakibul Islam, Rushdi Zahid Rusho, and Sheikh Md Rabiul Islam. Design and implementation of low cost smart syringe pump for telemedicine and health-care. In *2019 International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, pages 440–444. IEEE, 2019.
- [32] Mohannad Ismail, Jinwoo Yom, Christopher Jelesnianski, Yeongjin Jang, and Changwoo Min. Vip: safeguard value invariant property for thwarting critical memory corruption attacks. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*, pages 1612–1626, 2021.
- [33] Sashidhar Jakkamsetti, Youngil Kim, Andrew Searles, and Gene Tsudik. Eilid: Execution integrity for low-end iot devices. In *2025 Design, Automation & Test in Europe Conference (DATE)*, pages 1–7. IEEE, 2025.
- [34] Mathai Joseph and Paritosh Pandya. Finding response times in a real-time system. *The Computer Journal*, 29(5):390–395, 1986.
- [35] Mustakimur Khandaker, Abu Naser, Wenqing Liu, Zhi Wang, Yajin Zhou, and Yueqiang Cheng. Adaptive call-site sensitive control flow integrity. In *2019 IEEE European Symposium on Security and Privacy (EuroS&P)*, pages 95–110. IEEE, 2019.
- [36] Mustakimur Rahman Khandaker, Wenqing Liu, Abu Naser, Zhi Wang, and Jie Yang. Origin-sensitive control flow integrity. In *28th USENIX Security Symposium (USENIX Security 19)*, pages 195–211, 2019.
- [37] Jean-François Lalande, Karine Heydemann, and Pascal Berthomé. Software countermeasures for control flow integrity of smart card c codes. In *European Symposium on Research in Computer Security*, pages 200–218. Springer, 2014.
- [38] Alexandra Lengert, Adam Ilyas Caulfield, and Ivan De Oliveira Nunes. Boosting device utilization in control flow auditing. *arXiv preprint arXiv:2603.02161*, 2026.
- [39] Hans Liljestrand, Thomas Nyman, Lachlan J Gunn, Jan-Erik Ekberg, and N Asokan. Pacstack: an authenticated call stack. In *USENIX Security Symposium*, pages 357–374, 2021.
- [40] Ziyi Lin, Jinku Li, Bowen Li, Haoyu Ma, Debin Gao, and Jianfeng Ma. Typesqueezer: When static recovery of function signatures for binary executables meets dynamic analysis. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, pages 2725–2739, 2023.
- [41] ARM Ltd. Armv8-m memory model and memory protection user guide. <https://developer.arm.com/documentation/107565/0101/Introduction/Memory-Protection-Unit--MPU->, Accessed: April 2026.
- [42] Tanmaya Mishra, Thidapat Chantem, and Ryan Gerdes. Survey of control-flow integrity techniques for real-time embedded systems. *ACM Transactions on Embedded Computing Systems (TECS)*, 21(4):1–32, 2022.
- [43] Muhammad Nouman Nafees, Neetesh Saxena, Alvaro Cardenas, Santiago Grijalva, and Pete Burnap. Smart grid cyber-physical situational awareness of complex operational technology attacks: A review. *ACM Computing Surveys*, 55(10):1–36, 2023.
- [44] Antonio Joia Neto and Ivan De Oliveira Nunes. Isc-flat: On the conflict between control flow attestation and real-time operations. In *2023 IEEE 29th Real-Time and Embedded Technology and Applications Symposium (RTAS)*, pages 133–146. IEEE, 2023.
- [45] Antonio Joia Neto, Norrathep Rattanavipanon, and Ivan De Oliveira Nunes. Pearts: Provable execution in real-time embedded systems. In *2025 IEEE Symposium on Security and Privacy (SP)*, pages 47–47. IEEE Computer Society, 2024.

- [46] Job Noorman, Jo Van Bulck, Jan Tobias Mühlberg, Frank Piessens, Pieter Maene, Bart Preneel, Ingrid Verbauwhede, Johannes Götzfried, Tilo Müller, and Felix Freiling. Sancus 2.0: A low-cost security architecture for iot devices. *ACM Transactions on Privacy and Security (TOPS)*, 20(3):1–33, 2017.
- [47] Ivan De Oliveira Nunes, Karim Eldefrawy, Norrathep Rattanavipanon, Michael Steiner, and Gene Tsudik. VRASED: A verified Hardware/Software Co-Design for remote attestation. In *28th USENIX Security Symposium (USENIX Security 19)*, pages 1429–1446, 2019.
- [48] Ivan De Oliveira Nunes, Karim Eldefrawy, Norrathep Rattanavipanon, and Gene Tsudik. APEX: A verified architecture for proofs of execution on remote devices under full software compromise. In *29th USENIX Security Symposium (USENIX Security 20)*, pages 771–788, 2020.
- [49] Thomas Nyman, Jan-Erik Ekberg, Lucas Davi, and N Asokan. Cfi care: Hardware-supported call and return enforcement for commercial microcontrollers. In *Research in Attacks, Intrusions, and Defenses: 20th International Symposium, RAID 2017, Atlanta, GA, USA, September 18–20, 2017, Proceedings*, pages 259–284. Springer, 2017.
- [50] Johannes Obermaier and Vincent Immler. The past, present, and future of physical security enclosures: from battery-backed monitoring to puf-based inherent security and beyond. *Journal of hardware and systems security*, 2(4):289–296, 2018.
- [51] James Pallister, Simon Hollis, and Jeremy Bennett. Github Repository for BEEBS, 2019.
- [52] RIOT. RIOT-OS. <https://www.riot-os.org/>.
- [53] Ryan Roemer, Erik Buchanan, Hovav Shacham, and Stefan Savage. Return-oriented programming: Systems, languages, and applications. *ACM Transactions on Information and System Security (TISSEC)*, 15(1):1–34, 2012.
- [54] Md Golam Saqlain, Mohammad Rejwan Uddin, Asif Mahmood, and Mahady Hasan. Iot-enabled syringe pump system: enhancing medication accuracy and optimizing drug delivery with real-time alert system. In *2023 26th International Conference on Computer and Information Technology (ICCIT)*, pages 1–5. IEEE, 2023.
- [55] Sseed-Studio. Temperature Sensor, June 2015.
- [56] Gabriele Serra, Pietro Fara, Giorgiomaria Cicero, Francesco Restuccia, and Alessandro Biondi. Pac-pl: Enabling control-flow integrity with pointer authentication in fpga soc platforms. In *2022 IEEE 28th Real-Time and Embedded Technology and Applications Symposium (RTAS)*, pages 241–253. IEEE, 2022.
- [57] Xi Tan and Ziming Zhao. Sherloc: Secure and holistic control-flow violation detection on embedded systems. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, pages 1332–1346, 2023.
- [58] SymPy Development Team. sympy, May 2024.
- [59] Liam Tyler and Ivan De Oliveira Nunes. Untrusted code compartmentalization for bare metal embedded devices. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 43(11):3419–3430, 2024.
- [60] Moshe Y Vardi. An automata-theoretic approach to linear temporal logic. In *Logics for concurrency: structure versus automata*, pages 238–266. Springer, 2005.
- [61] Theo Walker. OpenSyringePump, April 2022.
- [62] Robert J Walls, Nicholas F Brown, Thomas Le Baron, Craig A Shue, Hamed Okhravi, and Bryan C Ward. Control-flow integrity for real-time embedded systems. In *31st Euromicro Conference on Real-Time Systems (ECRTS 2019)*, pages 2–1. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2019.
- [63] Yujie Wang, Cailani Lemieux Mack, Xi Tan, Ning Zhang, Ziming Zhao, Sanjoy Baruah, and Bryan C Ward. Insectacide: Debugger-based holistic asynchronous cfi for embedded system. In *2024 IEEE 30th Real-Time and Embedded Technology and Applications Symposium (RTAS)*, pages 360–372. IEEE, 2024.
- [64] Mario Werner, Thomas Unterluggauer, David Schaffnerath, and Stefan Mangard. Sponge-based control-flow protection for iot devices. In *2018 IEEE European Symposium on Security and Privacy (EuroS&P)*, pages 214–226. IEEE, 2018.
- [65] Xilinx. Vivado design suite user guide, 2017.
- [66] Jie Zhou, Yufei Du, Zhuojia Shen, Lele Ma, John Criswell, and Robert J Walls. Silhouette: Efficient protected shadow stacks for embedded systems. In *29th USENIX Security Symposium (USENIX Security 20)*, pages 1219–1236, 2020.